

# Electrical Services Design

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## Engineering Technology

May, 2026

## Electrical Services Design (A05930)

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|---------------------|----------------------------|----------------------------|
| <b>Short Title:</b> | Electrical Services Design |                            |
| <b>Faculty</b>      | Science and Computing      |                            |
| <b>Department</b>   | Engineering Technology     |                            |
| <b>Cluster</b>      | Electrical Engineering     |                            |
| <b>Author</b>       | CLAIRE.FITZPATRICK         |                            |
| <b>Credits</b>      | 5                          | <b>Level:</b> Intermediate |

### Description of Module / Aims

The aim of this module is to provide the knowledge and skills necessary to design and layout the electrical services and systems for a modern building/ plant.

### Programmes

|           |                                                                                       |           |
|-----------|---------------------------------------------------------------------------------------|-----------|
|           | BEng (Hons) in Electrical Engineering (WD_ETRIC_B)                                    | 2 / 4 / M |
|           | BEng (Hons) in Electrical and Automation Engineering (International)<br>(WD_ETRIC_BI) | 2 / 4 / M |
| ELEC-0034 | BEng (Hons) in Sustainable Energy Engineering (WD_ESERG_B)                            | 2 / 4 / M |
|           | BEng (Hons) (WD_ETRIC_B)                                                              | 2 / 4 / M |
| ELEC-0034 | BEng in Electrical Engineering (WD_EELCT_D)                                           | 3 / 6 / M |

### Indicative Content

- Cables & cable sizing, switchboard & switchgear design
- Electrical Containment Systems
- Fire Alarm Systems
- Lightning protection & UPS systems
- Testing & Certification

### Learning Outcomes

*On successful completion of this module, a student will be able to:*

1. Describe the electrical systems that are found in modern buildings & plant.
2. Apply standard calculation techniques necessary for the design of electrical systems within a modern building.
3. Recognise and understand the comparative cost and operational benefits of different types of electrical services and systems within modern buildings.
4. Explain the procedures involved in designing the electrical services for a building/ plant.
5. Apply electrical services design techniques.

### Learning and Teaching Methods

- Lectures, presentations, reading and problem solving in class.
- Site visits and (on site) presentation of specialist equipment and topics
- Practical design Exercises
- Case study based on the design and layout of electrical services and systems

## Assessment Methods

|                           | Weighing | Outcomes Assessed |
|---------------------------|----------|-------------------|
| Final Written Examination | 60%      | 1,2               |
| Continuous Assessment     | 40%      |                   |
| Assignment                | 40%      | 3,4,5             |

## Assessment Criteria

70% -

**100%:** The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

**60% - 69%:** The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

**50% - 59%:** The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

**40% - 49%:** The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

**<40%:** The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

## Learning Modes

| Learning Mode        | F/T Hours | P/T Hours |
|----------------------|-----------|-----------|
| Lecture              | 36        |           |
| Practical            | 12        |           |
| Independent Learning | 87        |           |

## Essential Material

- Electro Tech Council of Ireland, I. *National Rules for Electrical Installations*. 4th ed.. UK: ., 2008.
- Porges, F. and B. Rigby. *Design of Electrical Services for Buildings*. 4th ed.. UK: ., 2005.

## Supplementary Material

- Cook, P. *Electrical Installations Designs*. 4th ed.. UK: Wiley, 2013.
- Henshaw, M. *Protection against Lightning*. UK: BSI, 2007.
- Kitcher, C. *Practical Guide to Inspection Testing 2013*. 4th ed.. UK: Routledge, 2015.
- Koffler, R. *The Power Protection Guide*. UK: Stillwater Publishing, 2006.

## Requested Resources

- ENGINEERING LAB: Electrical